IMC round 1

```
import pandas as pd
import numpy as np
import os
import sys
```

```
df1 =
 ↪  pd.read_csv('round-1-island-data-bottle/round-1-island-data-bottle/prices_round_1_day_-2
df2 =
 ↪  pd.read_csv('round-1-island-data-bottle/round-1-island-data-bottle/prices_round_1_day_-1
df3 =
 ↪  pd.read_csv('round-1-island-data-bottle/round-1-island-data-bottle/prices_round_1_day_0.
```

```
df_squid = pd.concat([df1[df1['product'] == 'SQUID_INK'],df2[df2['product']
 ↪  == 'SQUID_INK'],df3[df3['product'] ==
 ↪  'SQUID_INK']]).reset_index(drop=True)
df_kelp = pd.concat([df1[df1['product'] == 'KELP'],df2[df2['product'] ==
 ↪  'KELP'],df3[df3['product'] == 'KELP']]).reset_index(drop=True)
```

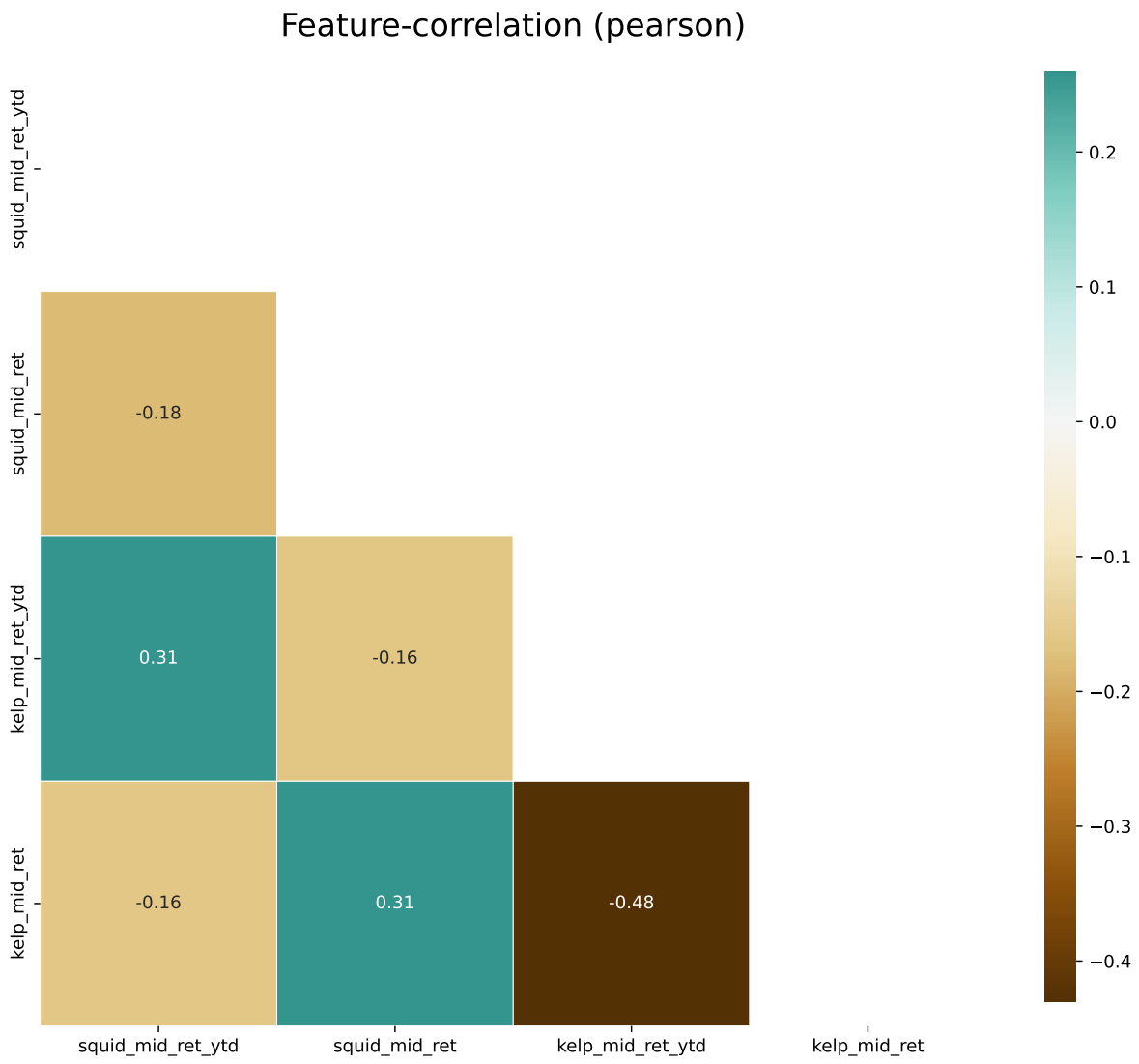
```
df_m =
 ↪  pd.merge(df_squid[['day','timestamp','mid_price']].rename(columns={'mid_price':'squid_mi
```

```
df_m['squid_mid_price_ytd'] = df_m['squid_mid_price'].shift(1)
df_m['kelp_mid_price_ytd'] = df_m['kelp_mid_price'].shift(1)
```

```
df_m['squid_mid_ret'] = df_m['squid_mid_price'].pct_change()
df_m['kelp_mid_ret'] = df_m['kelp_mid_price'].pct_change()
df_m['squid_mid_ret_ytd'] = df_m['squid_mid_ret'].shift(1)
df_m['kelp_mid_ret_ytd'] = df_m['kelp_mid_ret'].shift(1)
```

```
import klib as kl
```

```
kl.corr_plot(df_m[['squid_mid_ret_ytd','squid_mid_ret','kelp_mid_ret_ytd','kelp_mid_ret']])
```




```
df_m.set_index(['day','timestamp'], inplace=True)
```

```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.stats.diagnostic import lilliefors
import warnings
warnings.filterwarnings('ignore')
```

```
# Set the style for better visualizations
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("deep")
plt.rcParams['figure.figsize'] = [12, 8]
```

```
for col in ['squid_mid_ret','kelp_mid_ret']:
    # Basic descriptive statistics
    stats_df = pd.DataFrame({
        'Statistic': [
            'Count', 'Mean', 'Median', 'Std Dev', 'Skewness', 'Kurtosis',
            'Min', 'Max', '25%', '75%', 'Abs Mean', 'Positive Returns %'
        ],
        'Value': [
            df_m[col].count(),
            df_m[col].mean(),
            df_m[col].median(),
            df_m[col].std(),
            df_m[col].skew(),
            df_m[col].kurt(),
            df_m[col].min(),
            df_m[col].max(),
            df_m[col].quantile(0.25),
            df_m[col].quantile(0.75),
            df_m[col].abs().mean(),
            (df_m[col] > 0).mean() * 100
        ]
    })

    print(stats_df)

    # Reset index to plot with timestamp
    df_plot = df_m.copy().reset_index()
```

```
    df_plot['index'] = ((df_plot['day']+3)*1000000) + df_plot['timestamp']

    # Plotting the returns over time
    plt.figure(figsize=(14, 7))
    plt.plot(df_plot['index'], df_plot[col], linewidth=1.5)
    plt.title(f'{col} Returns Over Time', fontsize=16)
    plt.axhline(y=0, color='r', linestyle='-', alpha=0.3)
    plt.xlabel('Timestamp')
    plt.ylabel('Returns')
    plt.grid(True, alpha=0.3)
    plt.show()

    # Plot by day

    df_plot2 = df_plot.reset_index()

    plt.figure(figsize=(14, 10))
    for day in sorted(df_plot2['day'].unique()):
        day_data = df_plot2[df_plot2['day'] == day]
        plt.subplot(len(df_plot2['day'].unique()), 1, int(day) + 3)  # +3
↪   because days start at -2
        plt.plot(day_data['timestamp'], day_data[col], label=f'Day {day}')
        plt.axhline(y=0, color='r', linestyle='-', alpha=0.3)
        plt.title(f'Day {day}')
        plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()

    # Histogram and KDE
    plt.figure(figsize=(14, 6))

    plt.subplot(1, 2, 1)
    sns.histplot(df_m[col].dropna(), kde=True, bins=30)
    plt.axvline(x=0, color='r', linestyle='--', alpha=0.7)
    plt.title(f'Distribution of {col} Returns', fontsize=14)
    plt.xlabel('Returns')

    plt.subplot(1, 2, 2)
    sns.boxplot(y=df_m[col].dropna())
    plt.title(f'Box Plot of {col} Returns', fontsize=14)
    plt.ylabel('Returns')
```

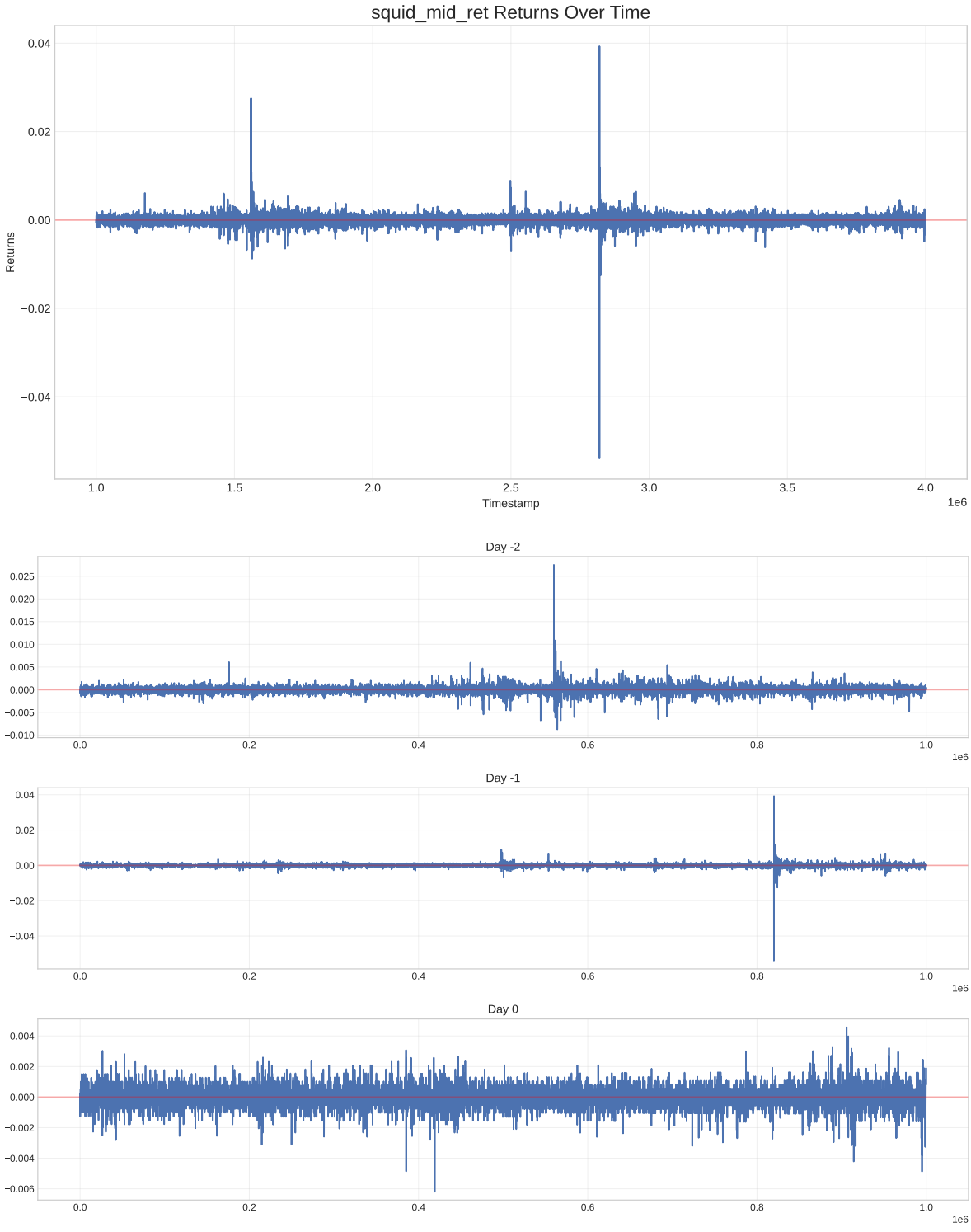
```
plt.tight_layout()
plt.show()

# QQ Plot to check for normality
plt.figure(figsize=(10, 6))
sm.qqplot(df_m[col].dropna(), line='s')
plt.title(f'QQ Plot of {col} Returns', fontsize=14)
plt.grid(True, alpha=0.3)
plt.show()

# Run normality tests
shapiro_test = stats.shapiro(df_m[col].dropna())
ks_test = lilliefors(df_m[col].dropna())
ad_test = stats.anderson(df_m[col].dropna(), dist='norm')

print(f"Normality Tests for {col} Returns:")
print(f"Shapiro-Wilk Test: statistic={shapiro_test[0]:.4f},
 ↪  p-value={shapiro_test[1]:.4f}")
print(f"Kolmogorov-Smirnov Test: statistic={ks_test[0]:.4f},
 ↪  p-value={ks_test[1]:.4f}")
print("Anderson-Darling Test:")
for i in range(len(ad_test.critical_values)):
    sl, cv = ad_test.significance_level[i], ad_test.critical_values[i]
    print(f"    {sl}%: {cv:.3f}, {'Reject' if ad_test.statistic > cv else
     ↪  'Accept'} null hypothesis")

# Autocorrelation analysis
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
plot_acf(df_m[col].dropna(), lags=20, ax=ax1)
ax1.set_title('Autocorrelation Function (ACF)', fontsize=14)

plot_pacf(df_m[col].dropna(), lags=20, ax=ax2)
ax2.set_title('Partial Autocorrelation Function (PACF)', fontsize=14)

plt.tight_layout()
plt.show()

# Calculate rolling statistics
window_sizes = [5, 10, 20]
df_rolling = df_m.reset_index()

plt.figure(figsize=(14, 10))
```

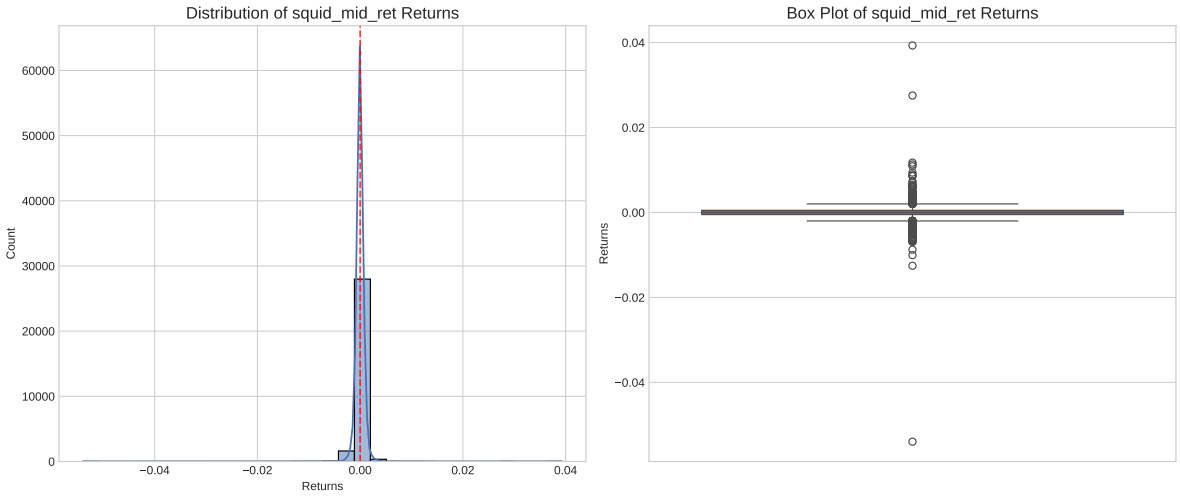
```
    # Rolling mean
    plt.subplot(3, 1, 1)
    plt.plot(df_rolling['timestamp'], df_rolling[col], label='Returns',
    ↪ alpha=0.5)
    for window in window_sizes:
        plt.plot(df_rolling['timestamp'],
                df_rolling[col].rolling(window=window).mean(),
                label=f'{window}-period MA')
    plt.title('Rolling Mean of SQUID_INK Returns', fontsize=14)
    plt.legend()
    plt.grid(True, alpha=0.3)

    # Rolling volatility (standard deviation)
    plt.subplot(3, 1, 2)
    for window in window_sizes:
        plt.plot(df_rolling['timestamp'],
                df_rolling[col].rolling(window=window).std(),
                label=f'{window}-period SD')
    plt.title('Rolling Volatility of SQUID_INK Returns', fontsize=14)
    plt.legend()
    plt.grid(True, alpha=0.3)

    # Rolling min/max
    plt.subplot(3, 1, 3)
    for window in window_sizes:
        plt.plot(df_rolling['timestamp'],
                df_rolling[col].rolling(window=window).min(),
                label=f'{window}-period Min', linestyle='--')
        plt.plot(df_rolling['timestamp'],
                df_rolling[col].rolling(window=window).max(),
                label=f'{window}-period Max', linestyle='-.')
    plt.title('Rolling Min/Max of SQUID_INK Returns', fontsize=14)
    plt.legend()
    plt.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.show()

    # Volatility clustering - plot squared returns
    plt.figure(figsize=(14, 6))
    plt.plot(df_rolling['timestamp'], df_rolling[col]**2)
```

```
    plt.title('Squared Returns (Volatility Clustering)', fontsize=14)
    plt.xlabel('Timestamp')
    plt.ylabel('Squared Returns')
    plt.grid(True, alpha=0.3)
    plt.show()

    # ACF of absolute returns to check volatility persistence
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
    plot_acf(df_m[col].dropna().abs(), lags=20, ax=ax1)
    ax1.set_title('ACF of Absolute Returns', fontsize=14)

    plot_acf(df_m[col].dropna()**2, lags=20, ax=ax2)
    ax2.set_title('ACF of Squared Returns', fontsize=14)

    plt.tight_layout()
    plt.show()

    # Correlation with Kelp returns
    plt.figure(figsize=(14, 6))
    plt.scatter(df_m['kelp_mid_ret'], df_m[col], alpha=0.6)
    plt.title('SQUID_INK Returns vs KELP Returns', fontsize=14)
    plt.xlabel('KELP Returns')
    plt.ylabel('SQUID_INK Returns')
    plt.axhline(y=0, color='r', linestyle='--', alpha=0.3)
    plt.axvline(x=0, color='r', linestyle='--', alpha=0.3)
    plt.grid(True, alpha=0.3)

    # Add correlation coefficient to the plot
    corr = df_m[[col, 'kelp_mid_ret']].corr().iloc[0, 1]
    plt.annotate(f'Correlation: {corr:.4f}',
                 xy=(0.05, 0.95), xycoords='axes fraction',
                 bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="b",
↪   lw=1))
    plt.show()

    # Correlation over time
    window = 10
    df_rolling['rolling_corr'] =
↪   df_rolling[col].rolling(window).corr(df_rolling['kelp_mid_ret'])

    plt.figure(figsize=(14, 6))
    plt.plot(df_rolling['timestamp'], df_rolling['rolling_corr'])
```

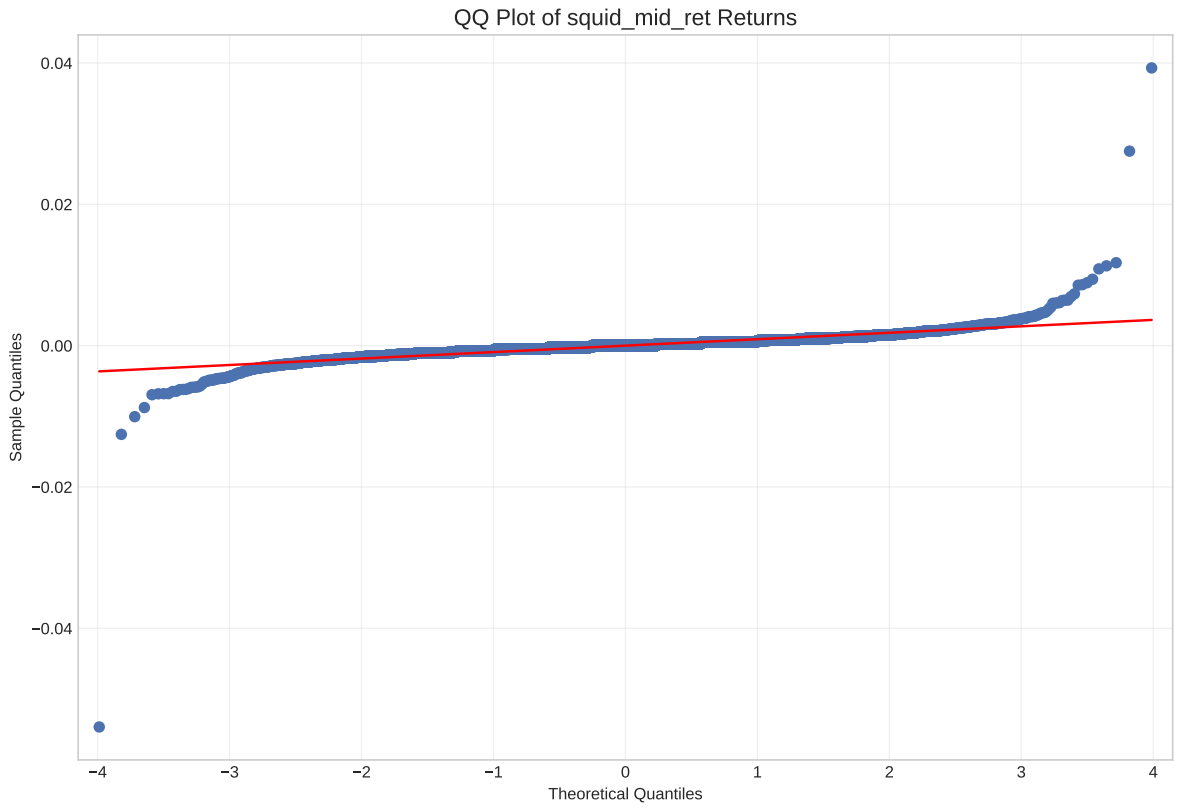
```
    plt.title(f'Rolling {window}-period Correlation between SQUID_INK and
↪   KELP Returns', fontsize=14)
    plt.xlabel('Timestamp')
    plt.ylabel('Correlation')
    plt.axhline(y=0, color='r', linestyle='--', alpha=0.3)
    plt.grid(True, alpha=0.3)
    plt.show()

    # Compare return distributions by day
    plt.figure(figsize=(14, 6))
    sns.violinplot(x='day', y=col, data=df_rolling)
    plt.title('Distribution of SQUID_INK Returns by Day', fontsize=14)
    plt.xlabel('Day')
    plt.ylabel('Returns')
    plt.grid(True, alpha=0.3)
    plt.show()
```

	Statistic	Value
0	Count	29999.000000
1	Mean	-0.000002
2	Median	0.000000
3	Std Dev	0.000912
4	Skewness	-3.308303
5	Kurtosis	559.090015
6	Min	-0.053963
7	Max	0.039301
8	25%	-0.000502
9	75%	0.000503
10	Abs Mean	0.000563
11	Positive Returns %	40.783333


squid_mid_ret Returns Over Time
Returns
0.04
0.02
0.00
−0.02
−0.04
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Timestamp
1e6
Day -2
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
0.0
0.2
0.4
0.6
0.8
1.0
1e6
Day -1
0.04
0.02
0.00
−0.02
−0.04
0.0
0.2
0.4
0.6
0.8
1.0
1e6
Day 0
0.004
0.002
0.000
−0.002
−0.004
−0.006
0.0
0.2
0.4
0.6
0.8
1.0
1e6

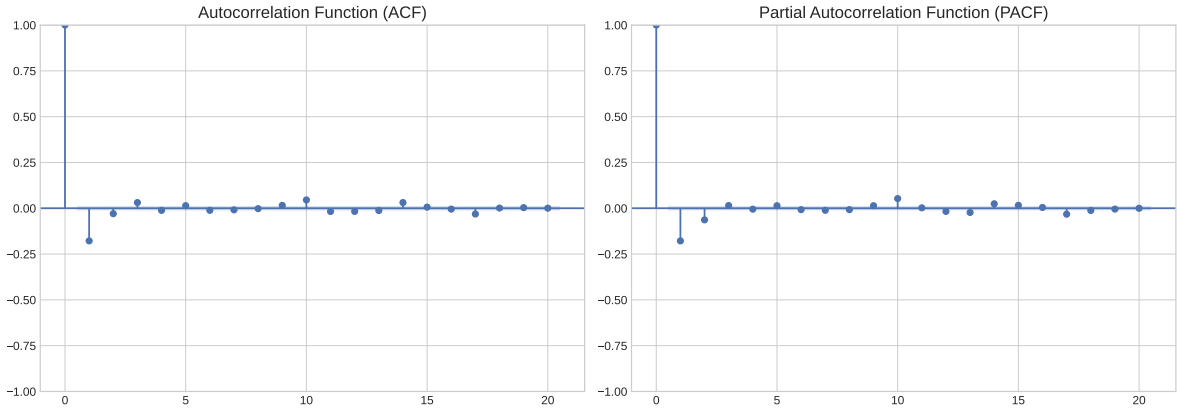


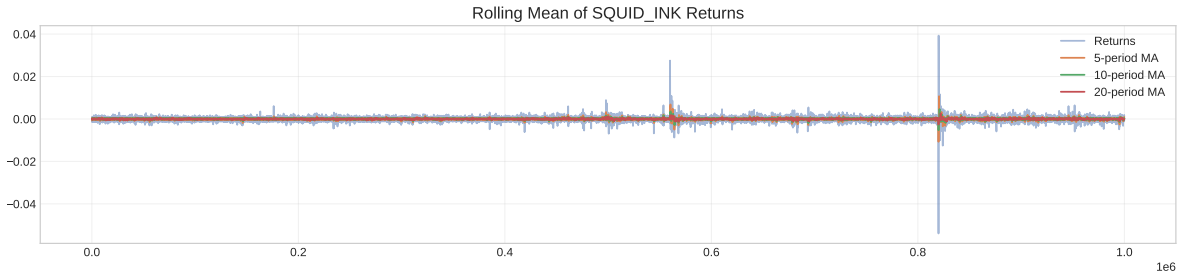

```
<Figure size 3000x1800 with 0 Axes>
```

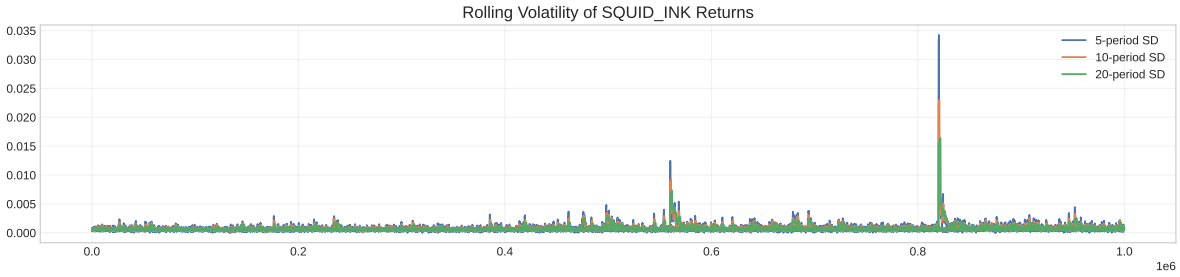


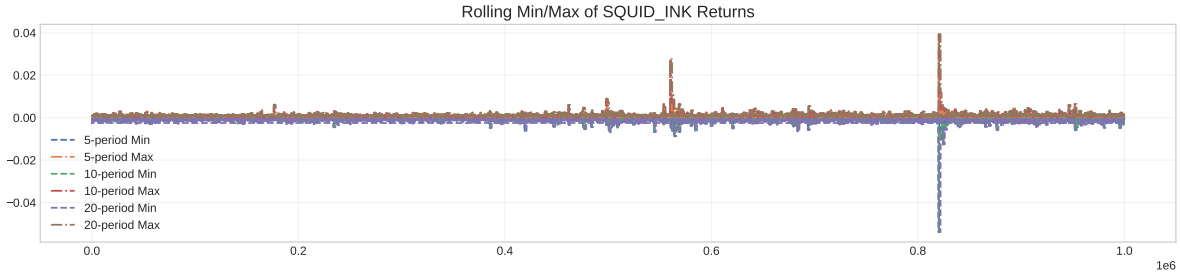

```
Normality Tests for squid_mid_ret Returns:
```

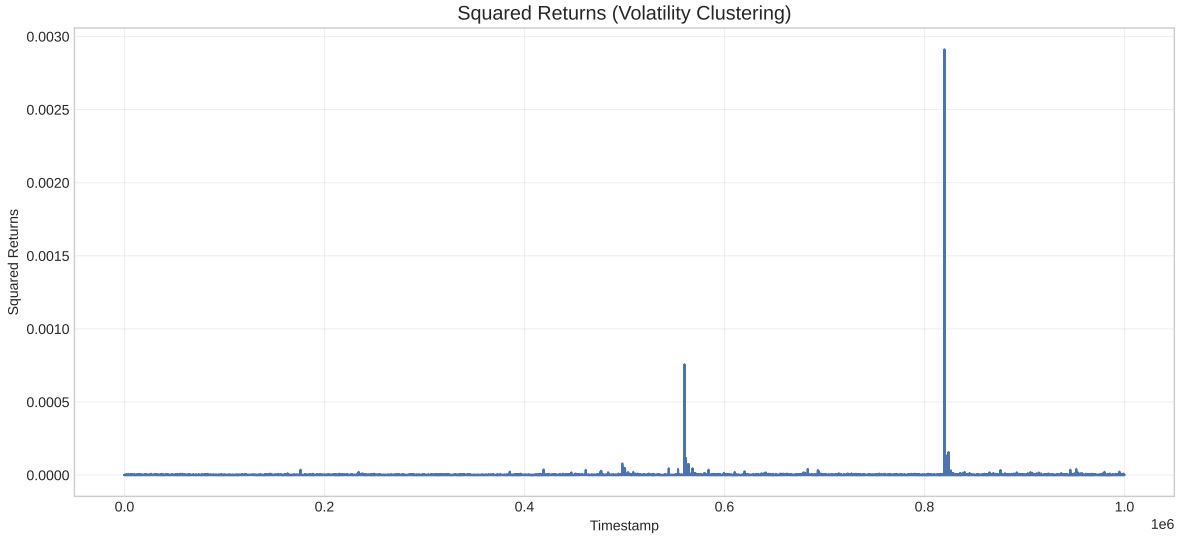
```
Shapiro-Wilk Test: statistic=0.7528, p-value=0.0000
Kolmogorov-Smirnov Test: statistic=0.1146, p-value=0.0010
Anderson-Darling Test:
    15.0%: 0.576, Reject null hypothesis
    10.0%: 0.656, Reject null hypothesis
    5.0%: 0.787, Reject null hypothesis
    2.5%: 0.918, Reject null hypothesis
    1.0%: 1.092, Reject null hypothesis
```

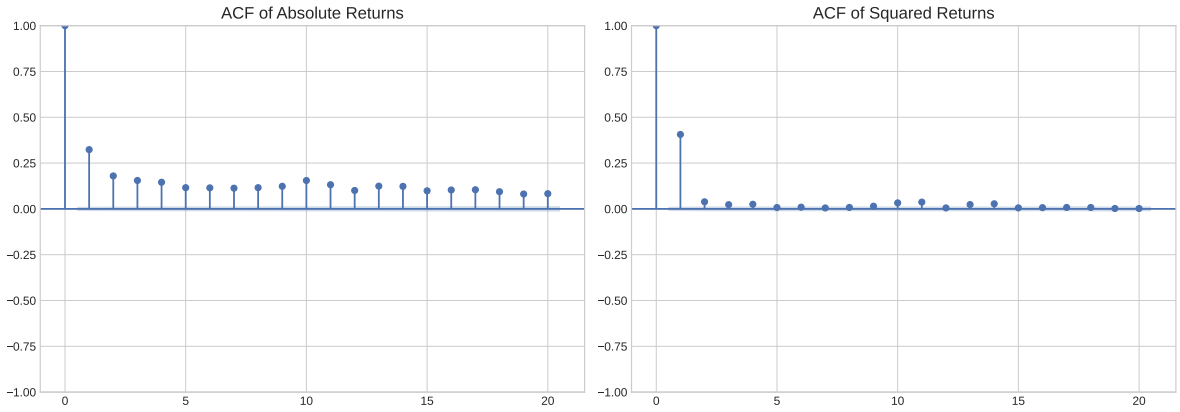


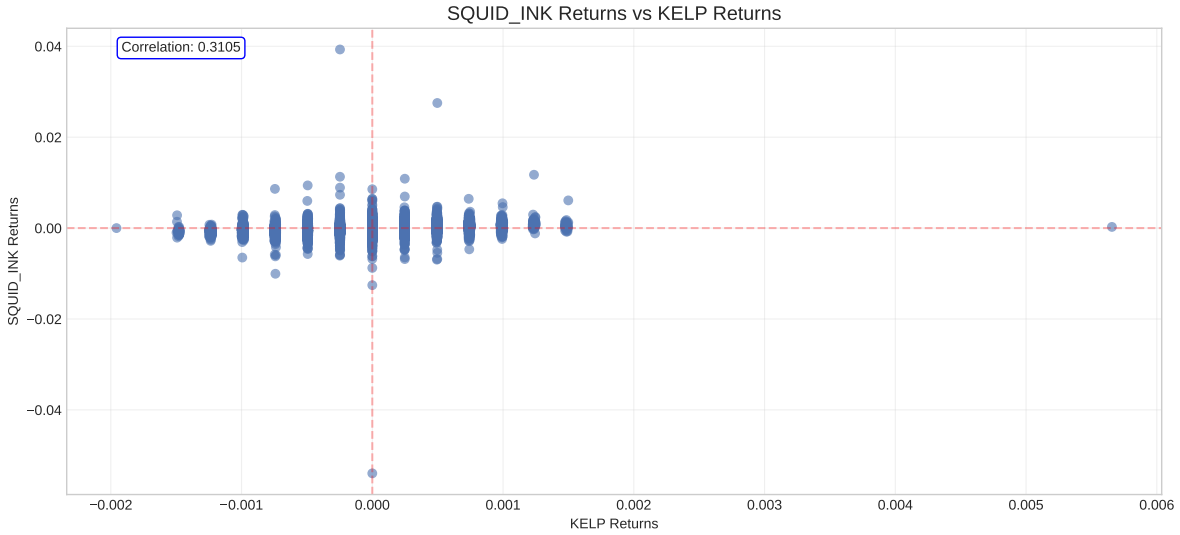

Rolling Mean of SQUID_INK Returns
Returns
5-period MA
10-period MA
20-period MA
0.04
0.02
0.00
−0.02
−0.04
0.0
0.2
0.4
0.6
0.8
1.0
1e6

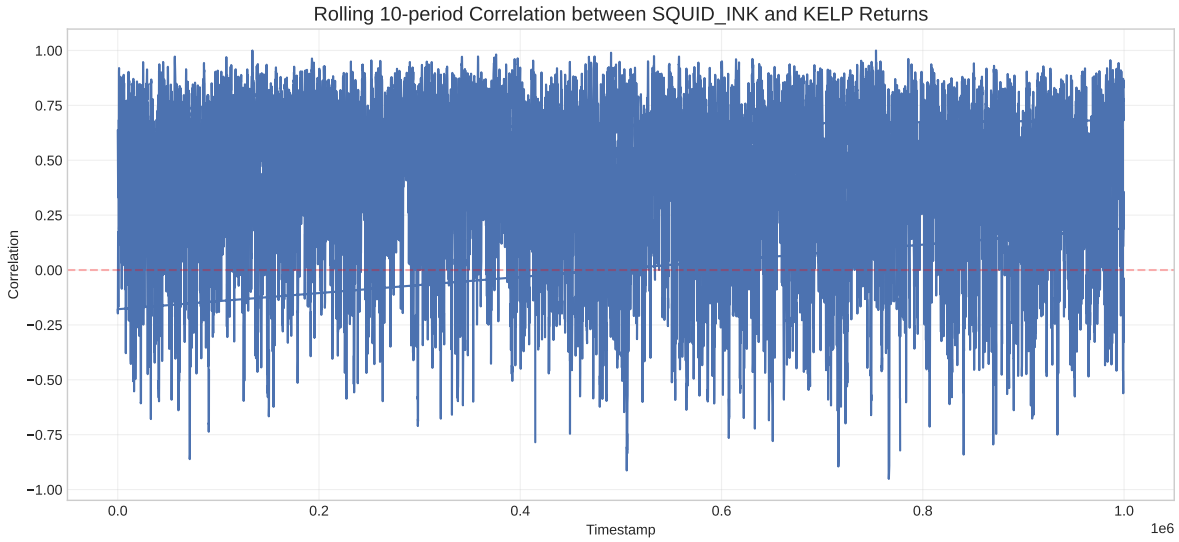

Rolling Volatility of SQUID_INK Returns
5-period SD
10-period SD
20-period SD
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0.0
0.2
0.4
0.6
0.8
1.0
1e6

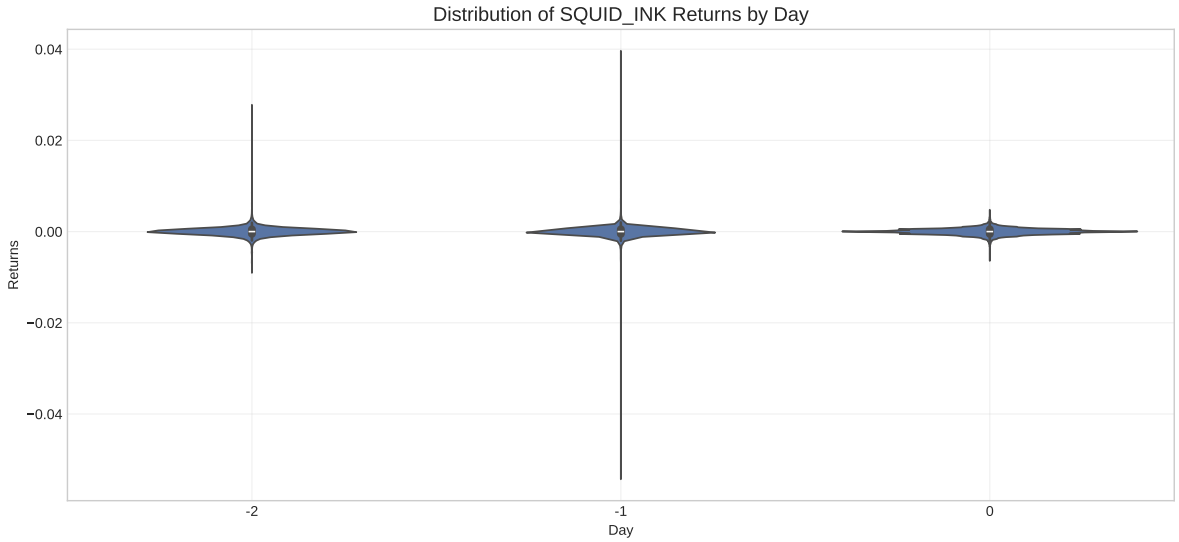

Rolling Min/Max of SQUID_INK Returns
5-period Min
5-period Max
10-period Min
10-period Max
20-period Min
20-period Max
0.04
0.02
0.00
−0.02
−0.04
0.0
0.2
0.4
0.6
0.8
1.0
1e6


Squared Returns (Volatility Clustering)
Squared Returns
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0.0
0.2
0.4
0.6
0.8
1.0
Timestamp
1e6


ACF of Absolute Returns
ACF of Squared Returns
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20

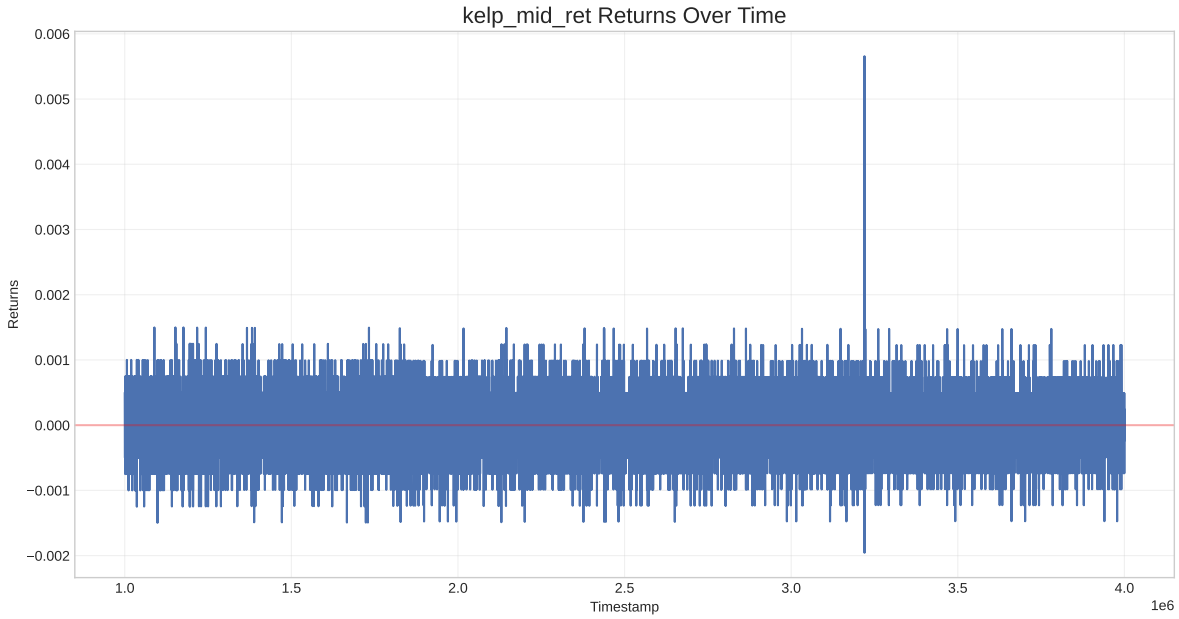

SQUID_INK Returns vs KELP Returns
Correlation: 0.3105
SQUID_INK Returns
KELP Returns
0.04
0.02
0.00
−0.02
−0.04
−0.002
−0.001
0.000
0.001
0.002
0.003
0.004
0.005
0.006

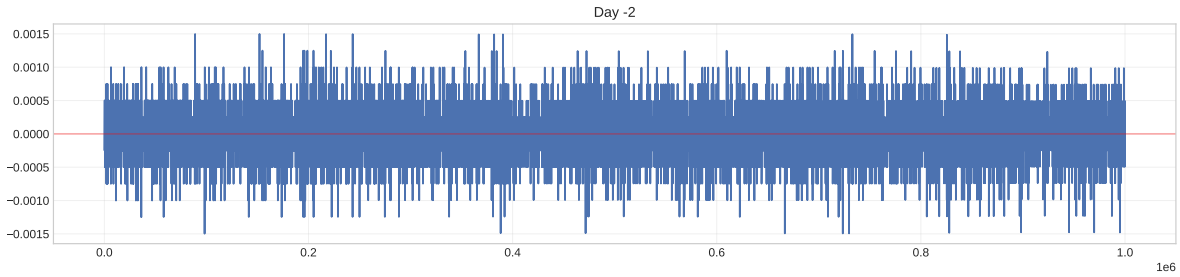


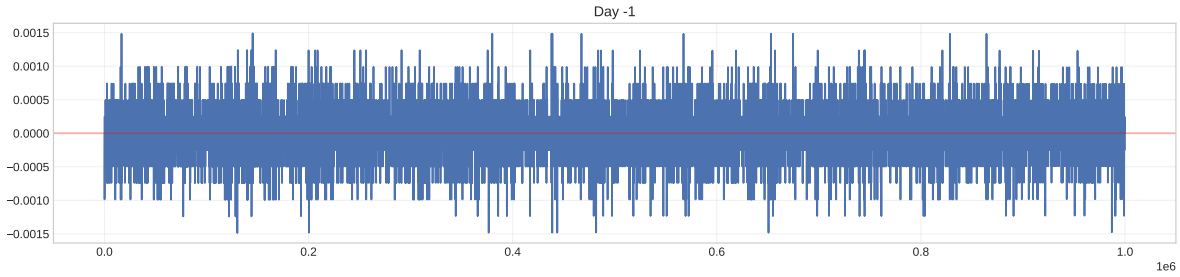


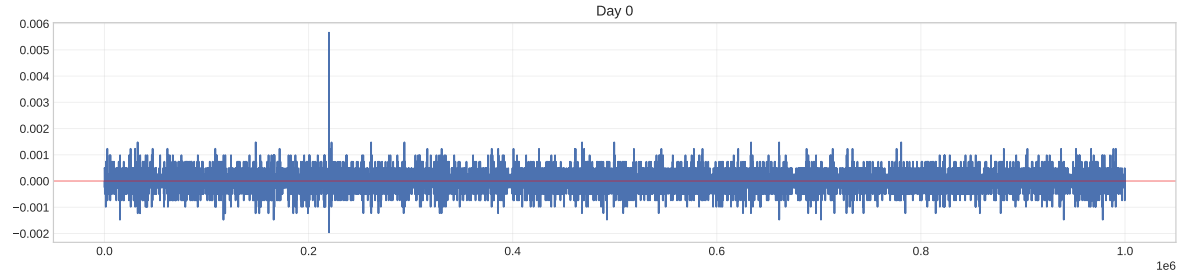
	Statistic	Value
0	Count	2.999900e+04
1	Mean	6.283120e-07
2	Median	0.000000e+00
3	Std Dev	3.862354e-04
4	Skewness	9.796893e-02
5	Kurtosis	2.386128e+00
6	Min	-1.956469e-03
7	Max	5.656665e-03
8	25%	-2.461841e-04

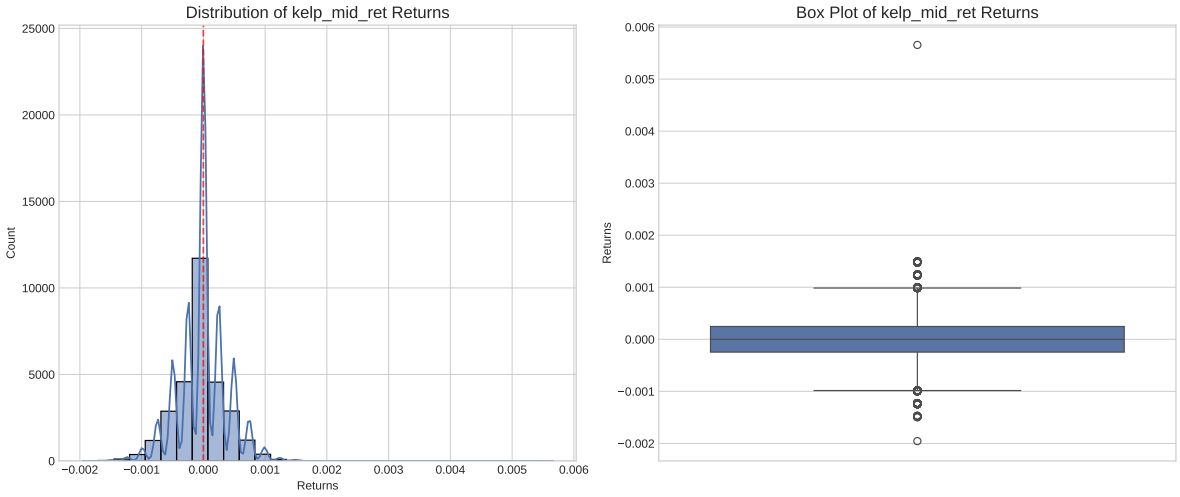
9	75%	2.463054e-04
10	Abs Mean	2.650698e-04
11	Positive Returns %	3.049333e+01



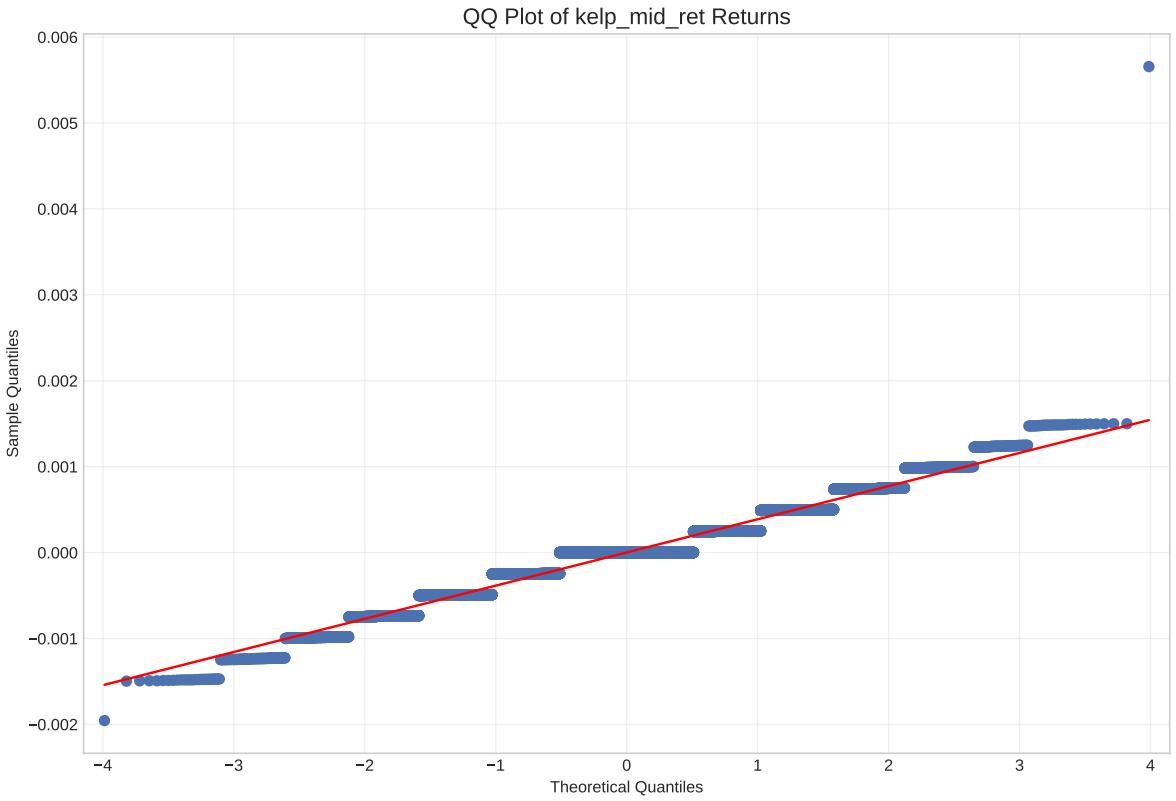




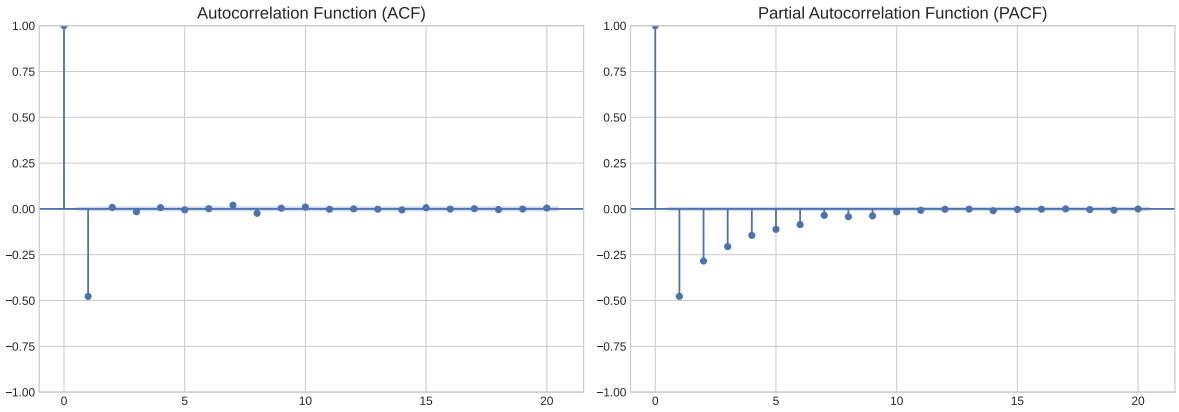


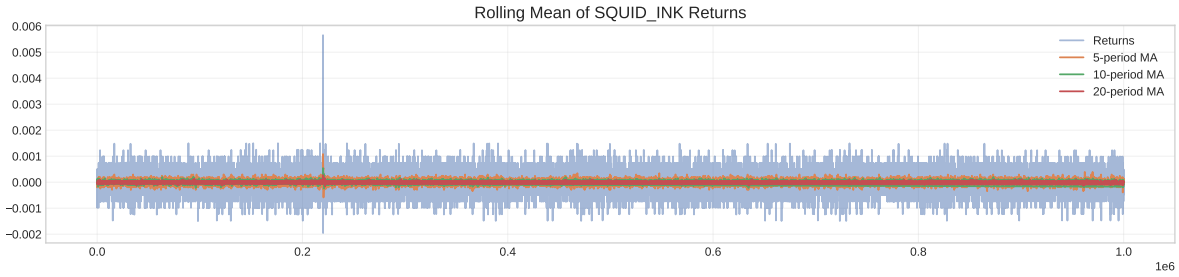


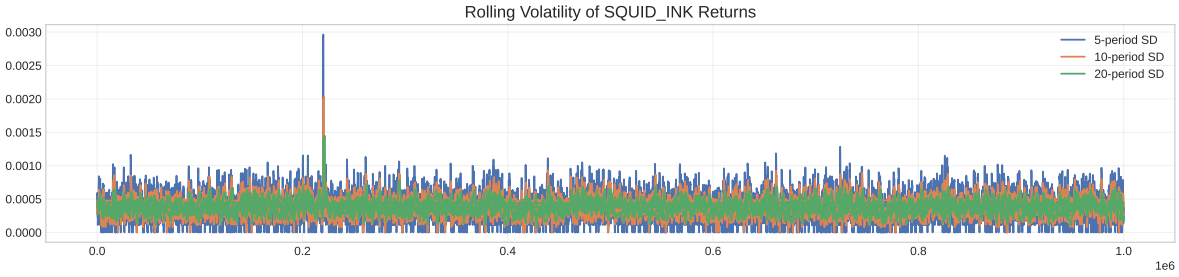
<Figure size 3000x1800 with 0 Axes>

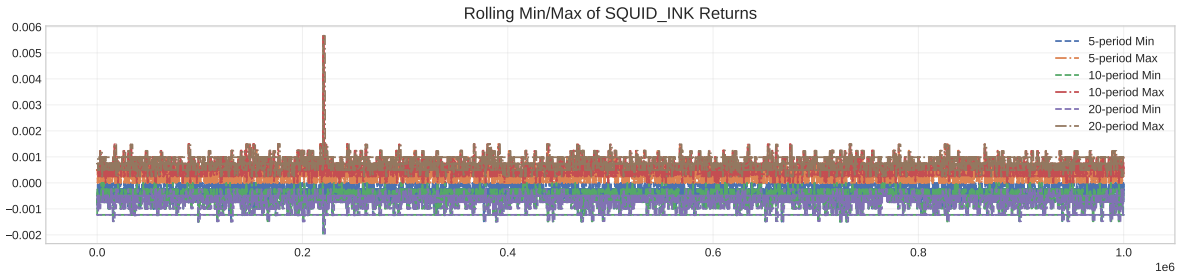


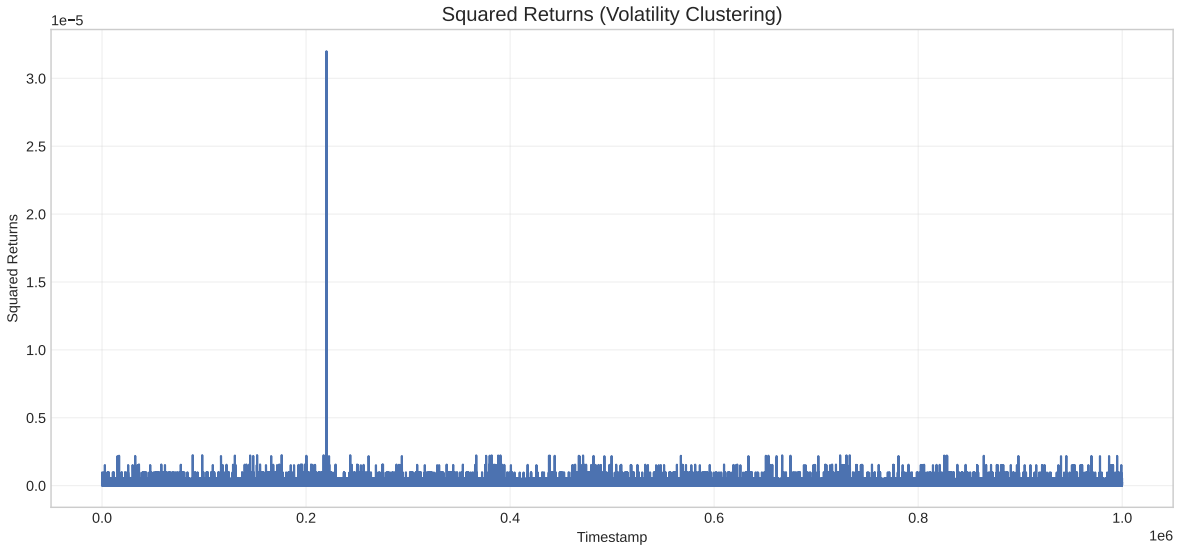

```
Normality Tests for kelp_mid_ret Returns:
Shapiro-Wilk Test: statistic=0.9440, p-value=0.0000
Kolmogorov-Smirnov Test: statistic=0.1957, p-value=0.0010
Anderson-Darling Test:
    15.0%: 0.576, Reject null hypothesis
    10.0%: 0.656, Reject null hypothesis
    5.0%: 0.787, Reject null hypothesis
    2.5%: 0.918, Reject null hypothesis
    1.0%: 1.092, Reject null hypothesis
```

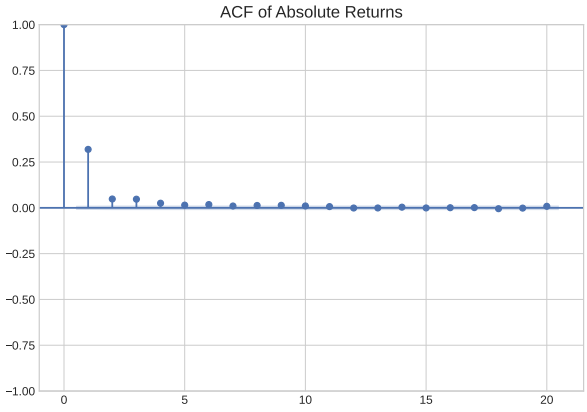

Autocorrelation Function (ACF)
Partial Autocorrelation Function (PACF)

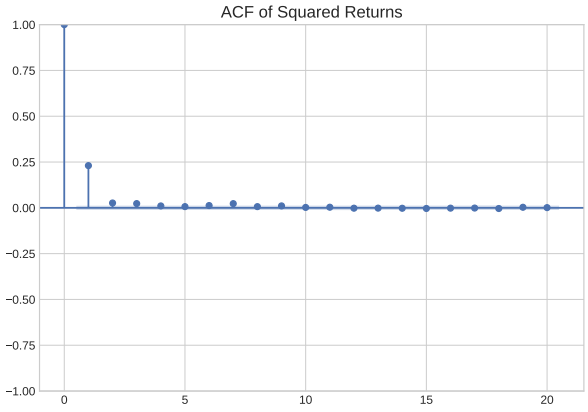

Rolling Mean of SQUID_INK Returns
Returns
5-period MA
10-period MA
20-period MA
1e6


Rolling Volatility of SQUID_INK Returns
5-period SD
10-period SD
20-period SD
1e6

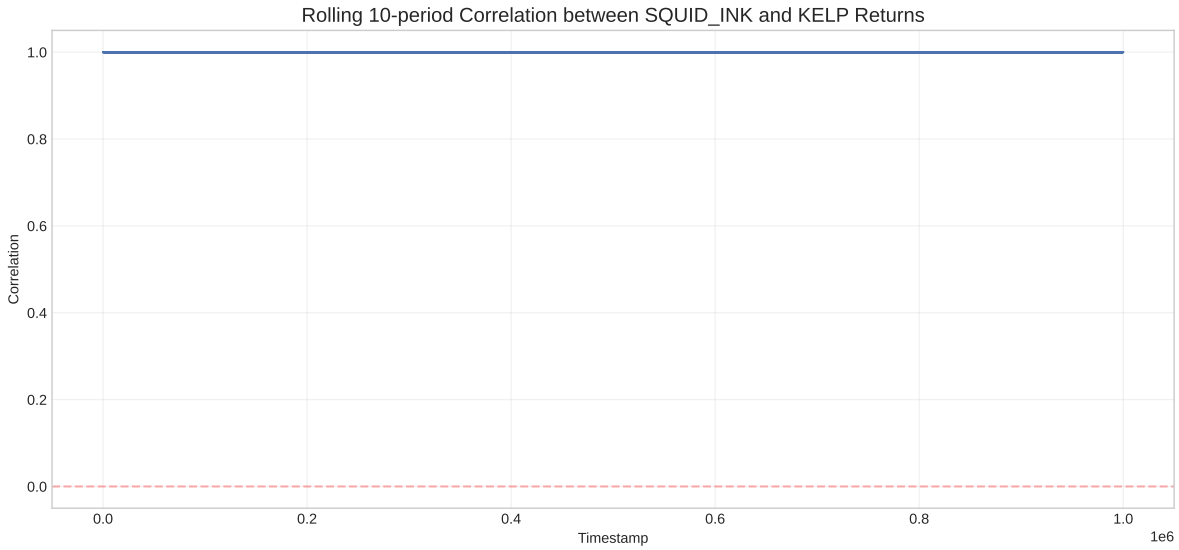

Rolling Min/Max of SQUID_INK Returns
5-period Min
5-period Max
10-period Min
10-period Max
20-period Min
20-period Max
1e6

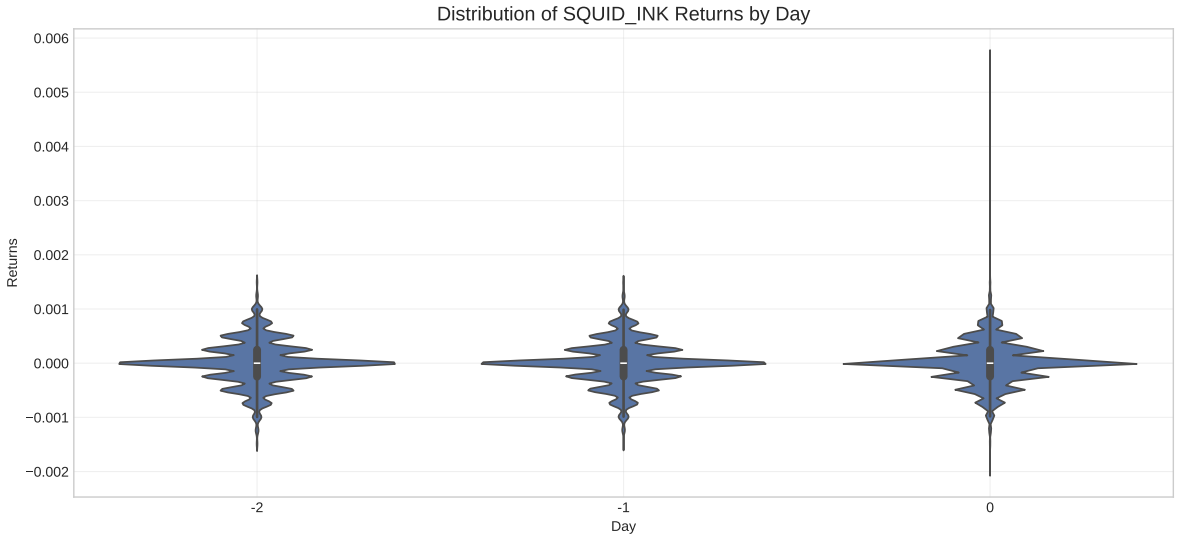

Squared Returns (Volatility Clustering)
1e−5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Squared Returns
0.0
0.2
0.4
0.6
0.8
1.0
Timestamp
1e6


ACF of Absolute Returns
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20


ACF of Squared Returns
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20


SQUID_INK Returns vs KELP Returns
Correlation: 1.0000
SQUID_INK Returns
KELP Returns
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−0.001
−0.002
−0.002
−0.001
0.000
0.001
0.002
0.003
0.004
0.005
0.006


Rolling 10-period Correlation between SQUID_INK and KELP Returns
Correlation
Timestamp
1e6
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


Distribution of SQUID_INK Returns by Day
Returns
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−0.001
−0.002
-2
-1
0
Day